

Chapter	Title	Total questions	Type
Ch 8	Journey Inside the Atom	6	Formative

This assessment is formative. Every question traces to a specific learning outcome, and each outcome is derived from an NCF competency this chapter develops. Rather than testing recall or asking whether a student “understood” a topic, each item asks for an observable demonstration — a response, a classification, a justification, or a product — of the thinking the lessons built. The questions are grouped by sections, and each shows the learning outcome it assesses and its cognitive demand, so you can see exactly what each question checks and why.

SECTION 1

8.1 Rediscovering the Roots of Atomic Theory

Q1 · Learning Outcome

Students can classify claims about natural phenomena as philosophical reasoning or experimentally grounded proposals and justify the distinction.

Cognitive Demand

Understanding

Kanada's concept of parmanu and the Greek idea of atomos both proposed that matter is made of indivisible particles. Which statement best describes what distinguishes Dalton's 1808 proposal from these earlier ideas?

- A. Dalton grounded his atomic proposal in experimental and chemical evidence rather than philosophical reasoning alone. ✓**
- B.** Dalton proposed that atoms combine into dyads and triads, whereas earlier thinkers claimed atoms never combine.
- C.** Dalton was the first thinker to suggest that matter is made of indivisible particles.
- D.** Dalton's atoms carry electrical charge, which the Greek atomos did not.

Answer section

Correct answer: A

What each choice reveals

- C** Confuses originality of the indivisibility idea with Dalton's actual contribution; ignores Kanada and the Greek tradition.
- B** Attributes the dyad/triad combining model specifically to Dalton, when it was Kanada's description; shows confusion between the two traditions.
- D** Imports a property from later atomic discoveries (Thomson's electron work) into Dalton's original model, where charge played no role.

SECTION 2

8.2 A Short Historical Journey Through Atomic Models / 8.2.2 Testing Thomson's model: The gold foil experiment

Q2 · Learning Outcome

Students can evaluate a scientific model by identifying the specific experimental observation that falsifies it and deriving a revised model from the evidence.

Cognitive Demand

Analysis

In the gold foil experiment, Geiger and Marsden observed that most alpha particles passed through the gold foil undeflected, some were deflected at large angles, and a very small number bounced almost straight back. Thomson's model predicted only slight deflections.

- (a)** Identify the single observation from the gold foil experiment that is sufficient to rule out Thomson's plum-pudding model, and explain precisely why that observation is incompatible with a uniformly distributed positive charge.
- (b)** State Rutherford's three conclusions from the experiment and link each conclusion to the specific observation that supports it.

Answer section

Look for

- Identifies sharp backscattering (very small number of alpha particles bouncing back) as the falsifying observation, not merely 'large deflections'.
- Explains that a diffuse positive charge cannot exert the concentrated repulsive force needed to reverse a fast, massive alpha particle's direction.
- States all three Rutherford conclusions: most of the atom is empty space; positive charge and most mass are concentrated in a small nucleus; electrons orbit the nucleus.
- Correctly links: undeflected passage → empty space; large-angle deflection and backscattering → concentrated nucleus; orbital motion → electron positions.
- Avoids conflating the gold foil result with Thomson's prediction — the response distinguishes what was predicted from what was observed.

SECTION 3

8.2.3 Bohr's model of the atom

Q3 · Learning Outcome

Students can represent energy transitions between quantised levels using a model and assess whether a postulate resolves the problem it was introduced to solve.

Cognitive Demand

Understanding

Rutherford's model predicted that orbiting electrons should continuously lose energy and spiral into the nucleus. Bohr resolved this by proposing stationary states. Which statement correctly describes what Bohr's stationary state postulate says about an electron in such a state?

- A. The electron can only exist in the K shell, where it is too close to the nucleus to radiate energy effectively.
- B. The electron is stationary — it has stopped moving — so it does not accelerate and therefore does not radiate energy.
- C. **The electron moves in a fixed orbit of definite energy and does not radiate energy while remaining in that orbit. ✓**
- D. The electron radiates a small, constant amount of energy that is replenished by the nucleus, so net energy loss is zero.

Answer section

Correct answer: C

What each choice reveals

- B Confuses 'stationary state' (fixed-energy orbit) with 'stationary electron' (not moving) — the most common linguistic misconception about Bohr's term.
- D Invents a compensating mechanism not present in Bohr's model; reveals a misunderstanding of the postulate as an energy-balance argument rather than a non-radiation assertion.
- A Incorrectly restricts the stationary-state condition to the K shell only; conflates the concept of lowest energy with the only allowed state.

SECTION 4

8.3 What Components Contribute to the Mass of an Atom? / 8.4 Symbols of Elements / 8.6 Mass Number

Q4 · Learning Outcome

Students can apply a symbolic notation system to determine the subatomic particle composition of an atom from given nuclear data.

Cognitive Demand

Application

An atom of phosphorus is represented as $^{31}_{15}\text{P}$.

- (a) State the number of protons, neutrons, and electrons in this atom. Show the calculation for neutron count.
- (b) A second atom has the symbol $^{27}_{13}\text{Al}$. Write the full nuclide notation for an atom of the same element that has 14 neutrons instead of 14 — wait, this atom has 14 neutrons. Calculate its mass number and write the corrected nuclide notation.
- (c) Explain in one sentence why the symbol for iron is Fe and not Ir, using the IUPAC rule for symbols derived from non-English names.

Answer section

Expected elements

- Phosphorus: protons = 15 (read directly from Z), electrons = 15 (atom is neutral so equals proton count), neutrons = $A - Z = 31 - 15 = 16$; calculation shown.
- For aluminium with 14 neutrons: $A = Z + \text{neutrons} = 13 + 14 = 27$; nuclide notation $^{27}_{13}\text{Al}$ — same as the given atom, demonstrating correct application of $A = Z + n$.
- Fe derives from the Latin name 'ferrum'; IUPAC allows symbols from Latin, Greek, or German names when the element was named in those languages, so the symbol need not match the English name.

SECTION 5

8.7 How Are Electrons Distributed in Different Energy Levels?

Q5 • Learning Outcome

Students can apply a rule-based system to construct the electron-shell distribution of an atom and extract the number of valence electrons from the result.

Cognitive Demand

Application

Use the Bohr-Bury shell-filling rules to answer the following.

- (a) Write the electronic configuration of sulfur ($Z = 16$) and silicon ($Z = 14$). Show the distribution across K, L, and M shells.
- (b) State the number of valence electrons for each element.
- (c) A student writes the configuration of chlorine ($Z = 17$) as 2, 8, 7. A second student writes it as 2, 9, 6. Identify which is correct and explain the error in the incorrect configuration.

Answer section

Expected elements

- Sulfur ($Z = 16$): K = 2, L = 8, M = 6; configuration 2, 8, 6; valence electrons = 6.
- Silicon ($Z = 14$): K = 2, L = 8, M = 4; configuration 2, 8, 4; valence electrons = 4.
- Correct configuration for chlorine is 2, 8, 7 (first student). The second student's error is placing 9 electrons in the L shell, which violates the $2n^2$ rule: for $n = 2$, maximum is $2(2^2) = 8$. Once L is full at 8, the remaining electrons go into M.

SECTION 6

8.8 Combining Capacity of an Atom: Valency

Q6 • Learning Outcome

Students can determine the combining capacity of an atom by reasoning from its electron configuration to the pathway by which it achieves a complete outer shell.

Cognitive Demand

Understanding

Magnesium has the electronic configuration 2, 8, 2. Which statement correctly identifies its valency and the pathway by which it achieves a complete outer shell?

- A. Valency 0, because magnesium's total electron count (12) is an even number, making it stable.
- B. Valency 2, because magnesium loses its 2 outermost electrons, leaving the L shell (8 electrons) as the new outer shell. ✓**
- C. Valency 2, because magnesium shares 2 electrons with another atom, adding 2 to reach an octet in the M shell.
- D. Valency 6, because magnesium needs to gain 6 electrons to complete its outermost shell to 8.

Answer section

Correct answer: B

What each choice reveals

- D Applies the gaining pathway instead of the losing pathway; the student counts 6 electrons needed to fill M to 8 without recognising that losing 2 is far easier and is the actual pathway for metals with few valence electrons.
- C Correctly identifies valency as 2 but assigns the wrong pathway (sharing rather than losing); the student knows the number but not the mechanism.
- A Invents a stability rule based on even electron count — a fabricated heuristic with no basis in the octet rule or the section's content.

